

Calibration of Trace range robust probe for O2 measurment

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Tags: AE Calibration O2 sensor Trace range robust oxygen sensor

Category: Protocol

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Goal

Performing calibration of [Equipment - Robust probe for liquid O2 measurment](#)

Prerequisites and preparation

desired flask with GL14 connction

[Equipment - Robust probe for liquid O2 measurment](#)

PT100

[Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel \(Firesting 2\)](#) (or another firesting

Laptop

Steps

Step number	Step description
1	Build the calibration setup with PT100 ideally also in reactor, for robust probe use 3 mm BOLA fitting, for PT100 use 4 mm BOLA fitting The setup is identically build as the degassing setup used for the experiments Add disired amount of water (typically same volume as used for the following experiments) If PT100 cannot be placed inside the reactor place it as close as possible in the same water bath as the reactor
2	Fill a balloon with compressed air, connect it in the way (or as close as possible) as the degassing would be done for irradiation experiments with this setup and start puriging with air (make sure to have a cannula in place to allow for the air to bubble though. The stirring should be done as for the irradiaiton experiments.
3	Calibrate the oxygen sensor spot by clicking on the gear symbol in the row of the Oxygen measurment data. Click on upper calibration and follow the insturction
4	When the oxygen value is constant (typically 5 to 10 min) press the button calibrated. Wait shortly after to see, if the choosen data point was an outlier or not. If it was take a new point.
5	Remove the cannula and the ballon
6	Purge the tubing used for the argon purging for approx. 1 min with a high argon flow
7	Purge the flask with argon, like it would be done for an irradiation experiment
8	click on lower calibration and follow the instructions.

9	Wait till the measurement is constant and in a reasonable range (typically 30 to 60 min, depending on the setup), in doubt: wait longer
10	Save calibration - > Finish.
11	Save the file with the calibration procedure in the following manner: Instruments -> Firesting -> Save instrument configuration (the file name is "date_Instrument_Settings_O2_cap) -> Save.
12	For checking the system, start logging (to investigate, how the system behaves, possible to perform it during 1 h).

Linked resources

Equipment - [Firesting Fiber-Optic Oxygen Meter 2 Channel \(Firesting 2\)](#)

Equipment - [Robust probe for liquid O2 measurement](#)

Protocol - [Calibration of O2 gas sensor cap](#)



Unique eLabID: 20250826-d1f28c8973db71d3189f5bfb6c68d4da331fd4d0d
Link: <https://elab.water-splitting.org/database.php?mode=view&id=261>